



SCHOOL OF
ARTIFICIAL INTELLIGENCE
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Towards Intelligent Diabetes Care: A Multimodal Digital Twin for Blood Glucose Prediction and Personalized Intervention

B.Tech Major Project Zeroth Review (Semester VII)

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Domain: Digital Health & Machine Learning — Team: C 14

Introduction & Motivation

Domain Background: Type 2 Diabetes (T2D)

- ▶ Type 2 Diabetes represents a long-term metabolic condition characterized by escalating insulin resistance, impaired pancreatic beta-cell function, and acute postprandial blood sugar surges.
- ▶ Although nutritional and behavioral adjustments serve as the cornerstone for managing glycemic levels and enabling potential clinical remission, standardized, generalized dietary intervention programs frequently fall short.

Why Personalized Management Matters Now

- ▶ **Metabolic Heterogeneity:** Uniform meal consumption leads to highly variable Postprandial Glycemic Responses (PPGR) between subjects, heavily driven by baseline systemic biomarkers and unique gut microbiome dynamics.
- ▶ **Technology & AI Driver:** Modern continuous glucose sensors (CGMs) paired with consumer fitness trackers allow deep temporal architectures to execute real-time personalized predictions and interactive counterfactual scenarios.

Literature Review

Author(s) & Year	Title & Journal	Methodology & Algorithms	Datasets & Inputs	Results	Limitations
Arshad et al. (2026) [1]	<i>Beyond glucose monitoring: Multi-analyte biosensing and AI-integrated wearable platforms for next-generation diabetes management.</i> Sensors and Actuators Reports	Combined chemical and optical tracking tools with AI algorithms, virtual twin models, and automated feedback loops.	Continuous bodily fluid data (sweat and tears) measuring a combination of insulin, lactate, and glucose levels.	Monitoring several biological markers at the same time greatly improves forecasting accuracy and proactive treatment compared to standard glucose-only methods.	Currently restricted to early lab models. Major challenges include scaling up manufacturing, lack of extensive human trials, and potential data privacy flaws in cloud storage.
Olawade et al. (2026) [2]	<i>Digital twin paradigm in diabetes prediction and management.</i> Diabetes Research and Clinical Practice	Built individualized digital replicas using flexible AI systems that constantly refresh based on live patient data.	Merged inputs from CGMs, smartwatches, dietary records, and standard hospital lab tests.	Accurately projected glucose behaviors and future complications. Automated intervention loops increased the amount of time patients stayed in safe blood sugar zones by 6 to 11 percent.	Relies heavily on brief initial studies. Scaling this technology is difficult due to privacy worries, missing universal guidelines, and unequal AI accuracy across different demographics.

Literature Review

Author(s) & Year	Title & Journal	Methodology & Algorithms	Datasets & Inputs	Results	Limitations
Kiran et al. (2026) [3]	<i>A digital twin framework for predicting and simulating type 2 diabetes onset using retrospective lifestyle data.</i> Frontiers in Digital Health	Used Cox survival models and causal graphs (DAGs) to test "what-if" scenarios regarding lifestyle changes.	Historical data spanning 17 years from nearly 20,000 UK Biobank participants, including mental health, food intake, and daily activity metrics.	Achieved high reliability in forecasting disease onset. The study confirmed that psychological stress and loneliness quicken diabetes development, particularly when paired with junk food diets.	Dependent on participants' memories of their lifestyle habits instead of objective sensor data. The causal framework might overlook unrecorded environmental or genetic variables.
Shamanna et al. (2024) [4]	<i>Personalized nutrition in type 2 diabetes remission: application of digital twin technology for predictive glycemic control.</i> Frontiers in Endocrinology	Blended machine learning techniques (CatBoost, LSTMs, Random Forests) to forecast glucose peaks and deliver immediate dietary recommendations.	Live CGM readings merged with step counters, sleep data, manual food entries, and historical clinical lab reports.	Lowered overall HbA1c scores substantially. The program allowed 94 percent of users to quit their prescriptions, with almost three-quarters sustaining disease remission.	The system's success demands intense user dedication to tracking meals manually and wearing devices daily. Its effectiveness in lower-income populations remains unknown.

Literature Review

Author(s) & Year	Title & Journal	Methodology & Algorithms	Datasets & Inputs	Results	Limitations
Zhang et al. (2024) [5]	<i>A framework towards digital twins for type 2 diabetes.</i> Frontiers in Digital Health	Applied regularized regression models and graph-based network algorithms to map connections within a vast medical knowledge database.	The Arivale dataset, featuring over a thousand molecular blood markers and standard clinical tests tracked over several months.	Advanced biological indicators (like proteins and metabolites) forecasted 6-to-12-month health shifts far better than basic age or weight metrics, discovering novel cellular targets.	Because the participants were primarily healthy, the findings do not easily apply to severe diabetes cases. The small group size limits confidence in long-term predictions.

Research Gap

Through our synthesis of current literature, we identified three major limitations in existing solutions:

1. **Burden of Manual Diet Logging:** Most digital twin platforms (like Shamanna et al., 2024) depend heavily on users manually recording every meal and portion size. This creates friction, causing tracking fatigue, inconsistent usage, and missing data points.
2. **Omission of Gut Microbiome Dynamics:** Traditional glucose prediction tools overlook gut microbiome profiles, missing out on how unique microbial ecosystems influence individual carbohydrate metabolism and post-meal spikes.
3. **Absence of Causal Interventions:** Conventional machine learning setups act as "black box" predictors that show future trends without explaining cause-and-effect. They do not allow users to test "What-If" scenarios—such as evaluating the effect of a post-meal walk—before taking action.

Problem Statement & Scope

In-Scope Boundaries:

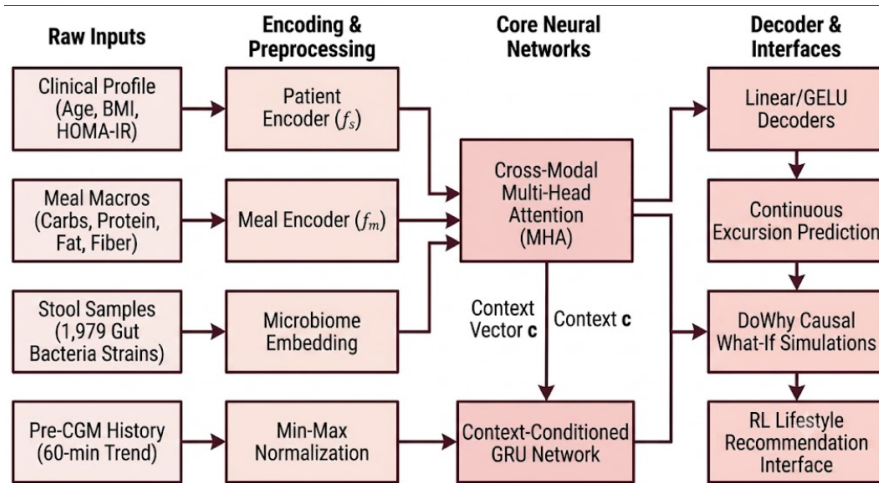
- ▶ **Glycemic Curve Forecasting:** Using GRU-based recurrent networks to model full 2–4 hour post-meal glucose trajectories, predicting key metrics like peak concentration, time-to-peak, and iAUC.
- ▶ **Microbiome Feature Compression:** Reducing the complexity of 1,979 binary gut bacteria indicators through PCA and Autoencoders to construct compact, meaningful microbial embeddings.
- ▶ **Causal Simulation Engine:** Designing a personalized Causal Directed Acyclic Graph (DAG) with DoWhy to run counterfactual tests estimating how dietary fiber or physical activity influences blood sugar.
- ▶ **Automated Vision Logging:** Training deep vision models to recognize dish types and estimate macronutrients directly from meal photos, removing the need for manual text input.

Objectives

The project is driven by the following three core objectives:

1. **Multimodal Glycemic Forecasting:** To develop a highly accurate predictive engine that forecasts post-meal glucose spikes within a **15% error margin** by integrating continuous glucose monitoring (CGM) time-series, wearable activity data, and individualized gut microbiome profiles.
2. **Dynamic Causal Simulation:** To construct a personalized "What-If" simulator that allows patients to dynamically evaluate the quantitative impact of counterfactual lifestyle interventions (e.g., altering meal macronutrients or scheduling post-meal walks) before execution.
3. **Frictionless Logging & RL Coaching:** To design an intelligent, computer-vision-assisted dietary logging tool aimed at eliminating patient logging fatigue, paired with a reinforcement learning coach to proactively maximize the patient's Time in Range (TIR: 70–180 mg/dL).

Proposed Methodology / Framework



Techniques & Tools (Software & Libraries)

The Digital Twin framework is implemented using the following software tools and libraries:

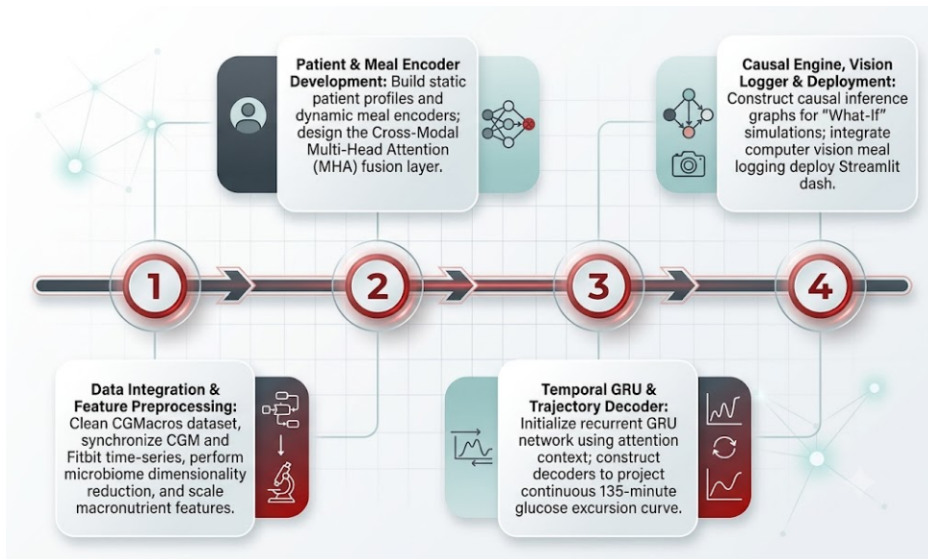
- ▶ **Data & Math Operations:** `pandas` and `numpy` for preprocessing; `scipy` for trapezoidal integration of glycemic area under the curve (iAUC).
- ▶ **Microbiome Embeddings:** `scikit-learn` (PCA / UMAP) and `PyTorch-based encoders` for compressing 1,979 binary features.
- ▶ **Glycemic Forecasting:** `PyTorch` for Multi-Head Attention (MHA) and Gated Recurrent Units (GRUs); baseline comparison with `XGBoost` and `CatBoost`.
- ▶ **Causal Simulator:** `DoWhy` and `EconML` for causal directed acyclic graphs (DAGs) and counterfactual lifestyle effect estimation.
- ▶ **Computer Vision Logger:** `PyTorch` (`EfficientNet` and `Vision Transformers`) for pre/post-meal food image classification.
- ▶ **Deployment Dashboard:** `Streamlit` for building the interactive virtual twin dashboard with user-adjustable lifestyle sliders.

Industry Relevance

The proposed Digital Twin platform aligns with the following healthcare sectors and commercial opportunities:

- ▶ **Digital Therapeutics (DTx) & Preventative Care:** Directly targets the global demand for software-as-a-medical-device (SaMD) systems that assist in metabolic syndrome management without medication compounding.
- ▶ **Personalized Wellness Platforms:** Provides a high-value software layer for consumer health wearable users, translating raw continuous readings into predictive simulations.
- ▶ **Clinical Support Ecosystems:** Acts as a remote monitoring cockpit for endocrinologists and diabetes coaches, enabling proactive adjustments to patient lifestyle guidelines.
- ▶ **Insurance Risk Optimization:** Offers opportunities for healthcare providers and insurers to design preventative health plans based on objective, twin-verified Time in Range (TIR) metrics.

Timeline



Team Roles

Team Member & Role	Sub-Task Ownership & Modules
Prem N <i>ML & Core Models</i> (CB.SC.U4AIE23241)	Responsible for gut microbiome dimensionality reduction (PCA/autoencoders), clinical variable integration, and optimizing the PyTorch Multimodal Attention Fusion & GRU networks.
Abhishek Pandey <i>Data Eng & Backend</i> (CB.SC.U4AIE205)	Responsible for raw dataset cleaning (CGM, Fitbit steps), chronological alignment, backend database structures, and training baseline PPGR regression models (XGBoost/CatBoost).
Divagar <i>Frontend & Simulator</i> (CB.SC.U4AIE23223)	Responsible for Streamlit dashboard interface, computer vision meal logger (EfficientNet), building the DoWhy causal simulator, and deploying the RL coaching recommendations.

References I

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- [4] P. Shamanna, S. Joshi, M. Thajudeen, L. Shah, T. Poon, M. Mohamed, and J. Mohammed, “Personalized nutrition in type 2 diabetes remission: application of digital twin technology for predictive glycemic control,” *Frontiers in Endocrinology*, vol. 15, p. 1485464, Nov. 2024.
- [5] Y. Zhang, G. Qin, B. Aguilar, N. Rappaport, J. T. Yurkovich, L. Pflieger, S. Huang, L. Hood, and I. Shmulevich, “A framework towards digital twins for type 2 diabetes,” *Frontiers in Digital Health*, vol. 6, p. 1336050, Jan. 2024.
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